Presenter Guide — W3D5 · The Gammas

> **What this is.** A per-slide guide so whoever presents a slide has the full context: concepts from scratch, how to read each figure, how to interpret each number, what to say (with a ready delivery script), and what questions to expect. One block per slide. **A note on how to use it.** I put this together from the final deck, Andrea's review and our Discord discussion — it's a shared aid, not a script anyone has to follow word-for-word. Please correct anything I got wrong. Numbers are all checked against `pipeline/02`. **Source of truth:** the final deck `manuscript/slides/source-snapshots/The Gammas - NMA project.pdf` and `pipeline/02_canonical_analysis_and_slides.ipynb`. On-screen copy is English; the talk is in English.

Numbering — this guide matches the final deck

The final deck is **6 spoken pages + 5 backups**. **Cover and team are merged into page 1** (that's why the numbering dropped by one vs the old storyboard). This guide uses **the projected-deck numbering**:

Page / Slide	Content	Presenter
1 · Cover + Team	Title + "Meet the Gammas"	Valeria opens
2 · Introduction	The question + Pattern/Direction + Avery anchor	**Valeria**
3 · Methods	336 HCP · 78-feature fingerprint · benchmark vs 360 activation	**Arefeh**
4 · Results	Primary r=0.366 + B→A transfer 0.398 (scatter)	**Jaime**
5 · Robustness checks	Direction refined + activation 0.60 vs 0.37 + 2 guardrails	**Goutham**
6 · Conclusions	Survives / Refined / Unresolved + thanks	**Kerem**
7–11 · Backups	Divider + directional + validation + future + references	on demand

Five presenters, **one slide each (slides 2–6)**. Cover/team and backups take no speaking slot. Split locked on Discord (24 Jul); two tweaks vs the draft lines: Goutham takes the **whole** slide 5 (his guardrails lead into the 0.60/0.37 numbers they qualify), and Jaime takes the **whole** slide 4 (0.366 + the 0.398 transfer). The 0.398 is also printed on slide 6 under *Survives*, so Kerem lands it as the closing headline.

Common delivery rules (apply to EVERY slide)

From Andrea's review (she was clear about these): 1. **Define every concept as you name it** — FC, reconfiguration, segregation. Her most repeated point; Arefeh asked for it directly. 2. **Name the axes out loud before any number** from a figure: *"always make sure people know what they're looking at."* 3. **Less text, more visual.** Trim technical detail; cut material goes to Q&A. 4. **Rehearse to the minute.** Andrea offered a rehearsal round and Friday tutorial time. Let's pre-agree who fields which questions. 5. **It's about the process, not the finding:** *"you won't get any prize if the study is better or not."* Nobody is defending a discovery; we tell what we did and what we learned.

Language guardrails we agreed on — worth avoiding on any slide: - Not "we changed / the hypothesis evolved" → "we predicted two things; one held, one was refined". - Not "confidence interval" for the ±0.024 → "split SD" (spread across partitions). - Not "independent external validation" → "identity-disjoint same-HCP transfer". - Not "dFC" / "dynamic FC" / "true network dynamics" → it's **static** FC aggregated per condition. - Not "model A beats model B" (scoreboard) → "specificity check". - No causality, adaptive benefit, or connectivity-specific mechanism claims.

Slide 1 — Cover + Team · Valeria opens

On screen: *"Functional Connectivity Reconfiguration in N-back Working Memory"* · the five names · **The Gammas** · pod/TA line (Pod 884 "Ifrit Ras el Hanout" · Megapod Lotus · TA Andrea Buccellato · Project TA Azman Akhter). No figure, no speaking slot (shown while Valeria greets).

What to know: The title names the **measure**, not a mechanism. "Reconfiguration" here is literally a **difference between two condition-averaged FC matrices** — no temporal dynamics implied. Andrea asked for the **team name prominent**: the deck is archived and read by next year's cohort.

Known housekeeping (not science): - Stock photos and filler roles ("Actor/Doctor/…") — partly done; Arefeh, Valeria and Jaime have real photos, Kerem and Goutham don't yet. - **Personal email addresses** on a deck that gets archived publicly — worth finding and removing before submission.

Delivery: *"Hi, we're The Gammas. Our project is on functional connectivity reconfiguration in an N-back working-memory task."* Move to the Introduction (slide 2).

Slide 2 — Introduction: the question and the two predictions · Valeria

On screen (verbatim): - Title: *"Does load-related brain connectivity predict working-memory performance?"* - Anchor: *"Avery et al. 2020 predicted 2-back accuracy from task FC in the same dataset (r = 0.36). We asked whether the change between loads carries that signal."* - *"We predicted two things"* - **Pattern** — *"A 78-feature FC fingerprint of the 2-back − 0-back change predicts performance."* - **Direction** — *"Higher load shifts networks toward integration; larger shifts accompany better performance."* - *"A distributed pattern and a one-number direction are different claims."*

What to know (from scratch):

The N-back task. Two working-memory difficulty levels: **0-back** (easy — "press when you see this specific target image"; pure attention, almost everyone is near ceiling) and **2-back** (hard — "press when the current image matches the one two positions back"; you must hold and update a buffer). 2-back is where people **differ**, so it's the condition of interest.

The two predictions — the key point of this slide. Not one hypothesis, two, of different natures: - **Pattern** is **multivariate and sign-free**: give the 78 numbers together to a model and see if the set predicts. It doesn't require any one of them to go up or down. Analogy: a 78-biomarker panel — you don't care about each one alone, you care whether the set is diagnostic. - **Direction** is **a single signed number**: load must **lower** segregation (more integration), and that shift alone must predict who performs better. Analogy: "fever rises, and higher fever means worse prognosis" — one variable, one direction.

Why two and not one. The original proposal (Arefeh, 15 Jul) was **one sentence** that already held both claims. When the data came in, one held and one didn't. Splitting them isn't rewriting history — it's making explicit what they always were: **two independent tests**. This covers us if someone asks "wasn't it a single integration hypothesis?".

Why Avery 2020. It's the calibration anchor. It predicted `acc_2bk` from task FC, **same dataset**, at r = 0.36. Cite it so that when **our 0.366** appears on slide 4, the audience judges it against that published bar, not in a vacuum. **One citation here, no more** (Andrea's instruction).

Delivery (~1 min):

> *"Our question: does the way brain connectivity changes with memory load predict how well someone performs? A 2020 study by Avery already predicted 2-back accuracy from connectivity in this same dataset, at r = 0.36 — we asked whether the change between loads carries that signal. We made two*

> *different predictions. First, a pattern: a 78-feature fingerprint of the 2-back-minus-0-back change predicts performance — this doesn't need any particular direction, just that the combined pattern carries information. Second, a direction: higher load pushes networks toward integration, and people with bigger shifts perform better — that's a single signed number. These are different kinds of claims, so we test them separately."*

Transition: *"So how did we actually measure that? Over to the method."*

Likely questions: - *"Wasn't it one hypothesis?"* → "The original proposal was one sentence containing both a pattern and a directional claim; we report them as the two separate tests they always were." - *"Why 2-back, not 0-back, as the target?"* → "0-back is near ceiling — everyone gets it. 2-back is where people differ, so it's where there's something to predict."

Avoid: saying the hypothesis changed; a second citation.

Slide 3 — Methods: from scans to a held-out prediction · Arefeh

On screen (verbatim): - Title: *"We evaluated a 78-feature FC difference in held-out people"* - Definitions band (3 lines): - *"Functional connectivity = the Pearson correlation between the time courses of two brain regions. Each node is one of 360 Glasser ROIs."* - *"FC reconfiguration = how much each network pair changes its coupling when memory load goes from 0-back to 2-back."* - *"System segregation = (mean within-network FC − mean between-network FC) ÷ mean within-network FC, computed per participant per condition (Chan et al. 2014)."* - **1 · Cohort** — *"336 participants · HCP N-back working-memory task · 360 Glasser ROIs"* - **2 · Condition frames** — *"0-back and 2-back frames kept separate, with no temporal overlap"* - **Figure: three network matrices** (0-back FC, 2-back FC, and their difference). Caption: *"12 Cole-Anticevic networks · 12 within + 66 between = 78 values per person"* - **3 · Model** — *"StandardScaler + RidgeCV, fitted inside each training fold"* - **4 · Held-out evaluation** — *"repeated 5-fold CV · fixed holdout · 1000-permutation null · B→A transfer (301 → 100, identity-disjoint)"* - *"All splits hold out participants; "±" = split SD, not CI."*

What to know (from scratch) — the three definitions are yours to say out loud:

Functional connectivity (FC). The **Pearson correlation** between two regions' time courses. Take how region A's signal rises and falls over time and region B's; if they move together, high positive correlation; if unrelated, near zero. Each region is a **node**; there are **360** (Glasser atlas).

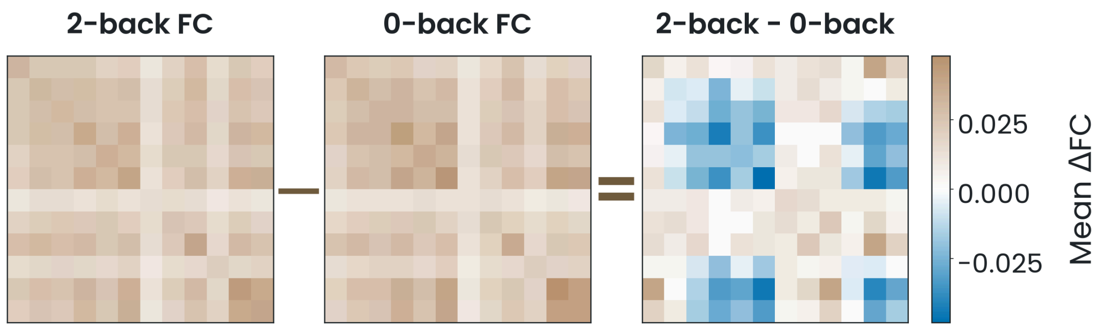
FC reconfiguration. How much each network pair changes its coupling from 0-back to 2-back. **2-back picture minus 0-back picture.** (Analogy: a cardiac stress test — what matters is the rest→exercise change, not either snapshot alone.)

System segregation (needs unpacking — 3 pieces): mean **within**-network connectivity, minus mean **between**-network connectivity, divided by the within (to normalize). One number per person per condition. **High = modular brain** (systems talking to themselves); **low = integrated** (networks mixing). Goutham will use this on slide 5 — remember "high modular, low integrated".

From 360×360 to 78. The full matrices would be >64,000 pairs — impossible with 336 people without heavy overfitting. So the 360 regions are grouped into **12 networks** (Cole-Anticevic atlas: visual, motor, cognitive control, default…) and averaged: **12** within-network values + **66** between-network (12×11/2 = 66 unique pairs) = **78 features** per person per condition.

The model (you all know this — don't dwell): ridge with standardization, both fit **inside** each training fold — never using test info. That's the hygiene that stops CV from inflating.

The 4 evaluations (each appears later): repeated 5-fold CV ×20 · fixed holdout · 1000-permutation null · B→A transfer with no shared identities.



The figure (read it this way — it's not self-evident): three 360×360 grids. Each cell = a region pair; colour = its correlation in that condition. Scale: blue (negative) → white (≈0) → tan (positive). Panel 1 = 2-back, panel 2 = 0-back, panel 3 = the difference.

⚠️ **Say this:** panels 1 and 2 share **one** scale; panel 3 has **its own**, ~10× tighter. Three equally intense-looking panels do **not** mean equal magnitudes — the difference panel is stretched to make a genuinely small change visible. Only panel 3 carries a colorbar.

Cut from the slide (available in Q&A): TR 0.72 s, 4 s HRF shift, 312 frames per condition, 2 runs per person, the 360×360 intermediates. Andrea: *"just say it's HCP, 336 participants, this is the task. That's it."* **Exception:** the **2-runs-per-person** design does need saying if anyone asks about the open (cross-run) squares on slide 5.

Delivery (~1 min):

> *"Quick definitions first. Functional connectivity is just the Pearson correlation between two brain regions' time courses — each region is one of 360 Glasser nodes. FC reconfiguration is how much each network pair changes its coupling from 0-back to 2-back. And system segregation — we'll need it later — is within-network minus between-network connectivity, normalized: high means a modular brain, low means an integrated one. Pipeline: 336 HCP participants doing the N-back. We compute a 360-by-360 correlation matrix per condition, then summarize it into 12 within- and 66 between-network values — 78 numbers per person. Naming the axes on this figure: three network matrices — 2-back, 0-back, and their difference. The first two share one colour scale; the third has its own, about ten times tighter, so the difference is small even though the colours look strong. Reconfiguration is that difference. We feed it to a ridge model, fitted strictly inside each training fold, and evaluate only on held-out people."*

Transition: *"So did this 78-feature difference actually predict anything? Yes — and it transferred."*

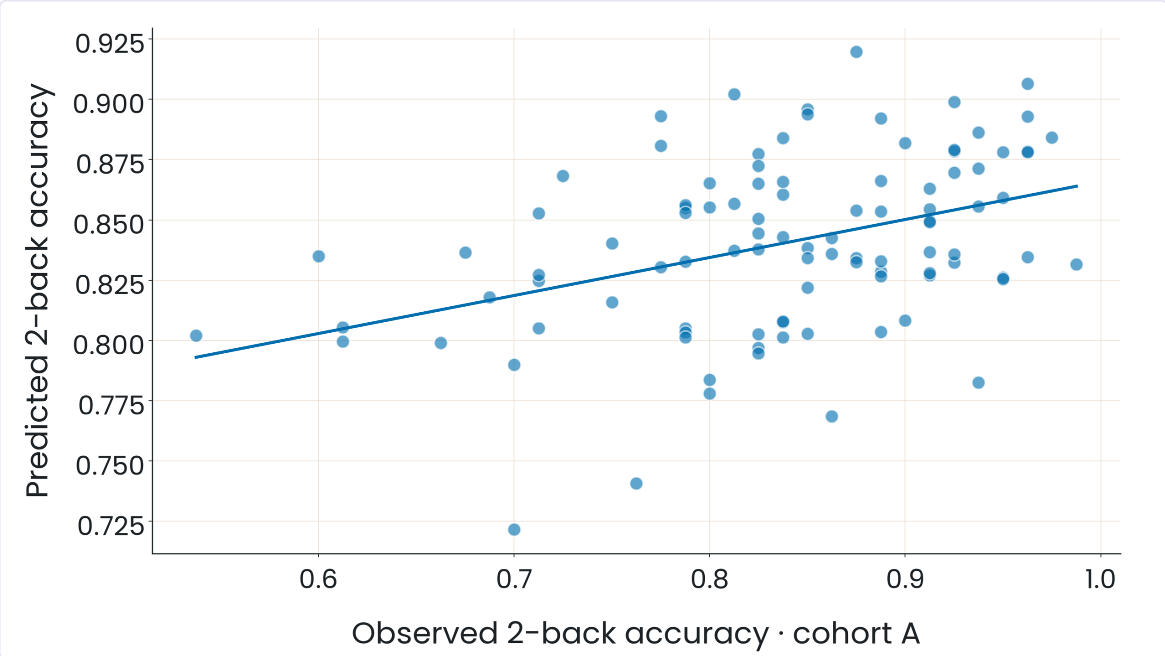
Likely questions: - *"Why 78 and not all pairs?"* → "64,000 edges for 336 people would overfit; the network summary is the standard dimensionality reduction." - *"Doesn't the task inflate correlations (coactivation)?"* → "Yes, task FC can reflect coactivation, not communication — it's a stated limitation; separating them is future work (backup slide 10, test 2)." - *"Acquisition parameters?"* → TR 0.72 s, HRF shift 4 s, 312 frames/condition, 2 runs — ready if asked.

Avoid: "dynamic FC"; implying the matrices show comparable magnitudes (different scales).

Slide 4 — Results: the pattern predicts and transfers · Jaime ✅

> **Your slide.** Goal in ~1 min: tell that **the pattern predicts and transfers**, make clear **how strong** each number is, and **don't overclaim**. When done, hand to **Goutham** (slide 5).

On screen (verbatim): - Title: *"The FC pattern predicted performance—and transferred to a separate cohort"* - **Primary repeated CV** *"r = 0.366 ± 0.024 · 336 participants · 78 FC features"* - **B→A transfer** *"r = 0.398; 95% CI [0.25, 0.53]"* · *"301 B → 100 A · 35 shared identities removed"* - Caveat: *"Identity-disjoint same-HCP transfer—not independent-site validation."* - Figure: scatter `identity-disjoint-transfer.png` (predicted vs observed, cohort A).



1 · The figure — name the axes BEFORE any number (Andrea's rule)

- **X** = each person's **actual** 2-back accuracy in **cohort A** (the test set).
- **Y** = the accuracy **predicted** by a model trained **only on cohort B** that never saw anyone in A.
- Each dot = one real person in A. The cloud **rises left to right** → there's signal. If the model were useless, it'd be a flat cloud.

2 · The two numbers — they measure different things, do NOT conflate them

	0.366 ± 0.024	**0.398 · CI [0.25, 0.53]**
What it is	**repeated-CV** mean (5-fold ×20 seeds)	**B→A transfer** (fixed model)
The ± / CI	**split SD** = how much it moves by partition	**bootstrap CI** = classic population uncertainty
Why they differ	model is **refit** each fold → measures *stability*, not a CI	model is **frozen** → you can resample A's 100 people

The **0.398 is your strongest number**: a model trained on one cohort, tested on **another**, with no shared identities (the 35 that appeared in both were removed).

3 · Calibration with Avery (why 0.36–0.40 is a good number)

0.366 and 0.398 land **right in the region of the published 0.36** from Avery in this same data — or a touch above. **Modest in absolute terms but on par with the literature.** The Project TA's success criterion was never a high R^2, but **clearly beating chance** — which the validation backup (slide 9) shows: 1000-permutation null $p \approx .001$ + fixed holdout 0.312.

4 · The mandatory caveat — say it yourself, don't wait to be asked

It's a transfer between **two cohorts of the same HCP**: stronger than reshuffling your own data, weaker than replicating at another scanner/population. Gradient: reshuffle-your-own-data < **same-HCP transfer (this)** < true external replication. Saying where you land = credibility (and it spares you the gotcha).

🎤 Your speech (English, ~65 s, ready to deliver)

> *"This is our main result. First, the axes: horizontal is each person's actual 2-back accuracy in cohort A — our test group; vertical is the accuracy our model predicted for them, from a model trained only on cohort B that never saw anyone in A. Each dot is one real participant, and the trend clearly rises.*
>
> *Two numbers, and they measure different things. Our primary effect is the repeated cross-validation correlation, r = 0.366 — and that ± 0.024 is a split standard deviation across partitions, how stable the number is, not a confidence interval. The stronger test: we froze a model trained on B only and applied it to A, after removing 35 people who appeared in both cohorts — so nobody is in training and test at once. That transfer gives r = 0.398, with a bootstrap 95% CI of 0.25 to 0.53.*
>
> *One honest caveat: this is an identity-disjoint transfer within the same HCP study — stronger than reshuffling your own data, but not an independent-site replication. Still, both numbers sit right on the 0.36 Avery reported in this same data, so this is a real, replicable brain-behavior effect — not a fluke."*

Handoff to Goutham: *"So the pattern held. But is that signal specifically about connectivity? That's what the next two checks tackle."*

Interpretation in depth (if they push)

- **Why split SD ≠ CI.** In repeated CV the model is retrained each fold, so the spread measures *partition sensitivity*, not population uncertainty. The classic CI is only legitimate for the **fixed** transfer model — that's why the CI goes with 0.398, not 0.366.
- **Why B→A and not A→B.** B is the larger cohort (336 vs 100): you train on the big sample and test on the small one; it's the more demanding, higher-powered generalization direction.
- **Circularity (same task).** Predicting N-back from N-back FC shares state variance. It's **partly controlled**: partialling out general ability (`acc_0bk`) the signal survives, and it's anchored in Avery's same-task design. Separating coactivation from connectivity = future work (slide 10, test 2).
- **Why does the number look low?** In brain–behavior prediction $r \approx 0.35$–0.40 is a real, publishable effect; the ceiling is also capped by the ceiling effect on `acc_2bk` (many people near 1.0).

Questions (likely + trap) — with ready answers

- *"What if it's overfitting or luck?"* → *"Backup: a 1000-permutation null gives p ≈ .001, and a fixed holdout of 67 never-seen people gives r = 0.312 — three procedures, same story."* (slide 9)
- *"Isn't 0.366 low?"* → *"For brain–behavior prediction it's a real, publishable effect — Avery got 0.36 in the same data. The bar was beating a permutation null, not a high R^2."*
- *"Why split SD and not a CI on 0.366?"* → *"The model is refit every fold, so that spread is partition sensitivity. We only claim a true CI for the fixed-model B→A transfer."*
- *"Isn't predicting N-back from N-back FC circular?"* → *"It's a same-task association — a stated limitation. Partialling out 0-back accuracy the signal survives, and we anchor it in Avery's same-task design; cleanly separating coactivation is future work."*
- *"Why train on B and test on A, not the reverse?"* → *"B is the larger cohort — 336 vs 100 — so we train on the bigger sample and test on the smaller, the stronger generalization direction."*

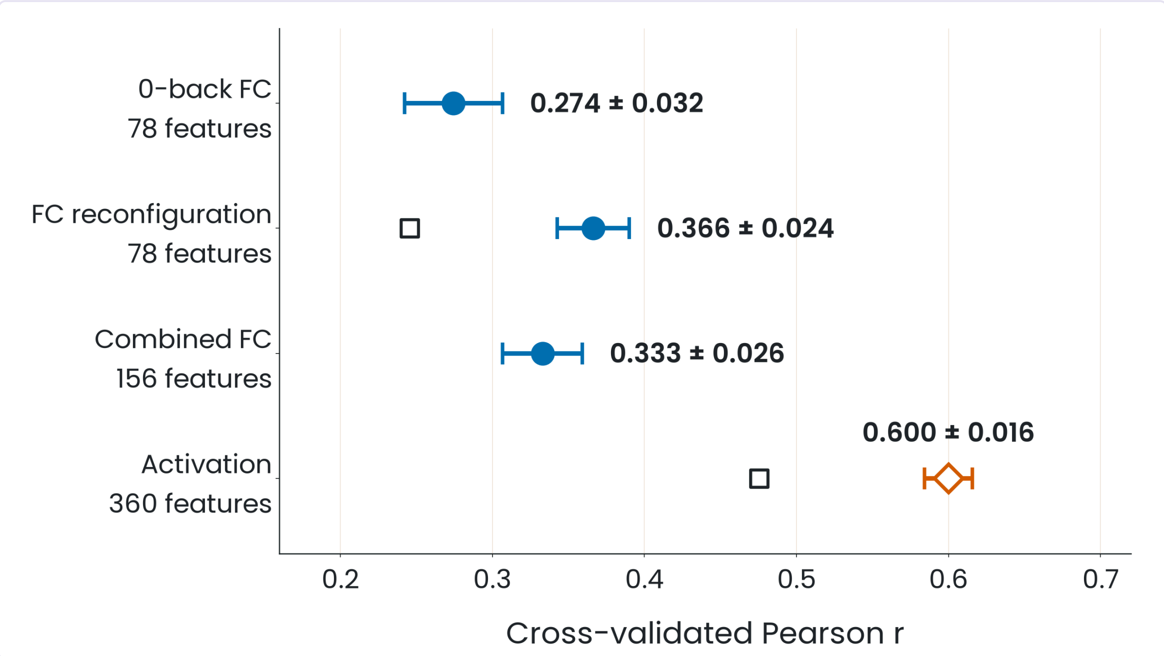
- *"Is this an external / independent validation?"* → *"No — identity-disjoint transfer within the same HCP study. Stronger than reshuffling, weaker than an independent-site replication."*

Avoid: "confidence interval" for 0.366 · "external / independent validation" for the transfer · saying the model "wins" anything (that's Goutham's slide) · "dynamic FC".

Delivery: ~1 min, one figure and two numbers. The only real risk is **rushing and merging the two numbers** — breathe between *"0.366, split SD"* and *"0.398, bootstrap CI"*. Rehearse it against the clock.

Slide 5 — Robustness checks: two checks that narrowed what we can claim · Goutham (the hinge)

On screen (verbatim): - Title: *"Two checks that narrowed what we can claim"* - *"Repeated-CV correlation (mean ± split SD across 20 partitions)"* - *"0-back FC 0.274 ± 0.032 · FC reconfiguration 0.366 ± 0.024"* - *"0-back + reconfiguration 0.333 ± 0.026 · Activation contrast 0.600 ± 0.016"* - **Direction, as predicted—but only at group level:** *"Segregation fell under load (0.3271 → 0.3035; Δ = −0.0236; p = 3.45 × 10^{-5}), yet larger shifts did not predict better performance (r = −0.105; p = .054)."* - *"Open squares = held-out cross-run generalization."* - Two caveat lines (large): - *"A specificity check, not a competition: post hoc, 360 activation features vs 78 FC features, and 0-back activation alone predicts as well (0.571), so the benchmark is not load-specific. FC reconfiguration stays the load-specific measure."* - *"Activation is a raw BOLD amplitude difference, not a GLM beta. Individual vascular reactivity (CVR) is uncontrolled, and this dataset carries no CVR proxy."* - **Figure:** `activation-robustness.png`.



Framing (say it this way): not "we lost to activation", but "we ran **two robustness checks**, and both made us more precise". Two beats: **direction** and **specificity**.

Check 1 — direction (answers the "Direction" prediction from slide 2): at group level, segregation fell 0.3271 → 0.3035; Δ = −0.0236; p = 3.45×10^{-5} (paired, n=336) — **essentially zero by chance**. The predicted direction **exists**. But does it predict an individual? The correlation between a person's segregation change and their accuracy is **r = −0.105, p = 0.054** — right at the significance border. So **no.**

> **The most transferable lesson of the talk:** an overwhelming group effect does **not** guarantee individual predictive value. Just as blood pressure rises with age across a population (rock-solid) but age alone doesn't diagnose a patient. Effect size ≠ individual diagnostic relevance.

Check 2 — specificity (the figure): dot-and-error-bar plot, horizontal, 4 rows. X = CV correlation, ~0.2 to 0.7. Each row = a different feature set, same model recipe: - **0-back FC** (78) → **0.274** - **FC reconfiguration** (78) → **0.366** (the star of slide 4) - **Combined** (156) → **0.333** — note, **lower** than reconfiguration alone: combining doesn't help, it dilutes. A hint that reconfiguration is already doing all the useful FC work. - **Activation** (360) → **0.600**, in orange/diamond — clearly above any FC.

Open squares over the reconfiguration and activation rows = an even stricter check (train on one run, predict on the other). By eye (**approximate, not printed numbers**): ~0.24 reconfig vs its 0.366; ~0.47 activation vs its 0.600. Honest message: cross-run replication is weaker for **both** measures.

Why it's NOT a fair competition (say all of it): 1. **360 vs 78** features, unmatched → more features almost always predict more, with no biological meaning. 2. **0-back activation alone already predicts 0.571** — nearly the 0.600 of the contrast. If the person just sitting in the easy condition already gives you almost all the prediction, the signal is **not about load**; it looks like a **stable trait**. 3. **CVR (cerebral vascular reactivity):** activation is raw BOLD amplitude (not a GLM beta), influenced by vascular factors (vessel stiffness, age, caffeine) unrelated to neural activity. This dataset can't measure or control it. FC, being a **correlation**, cancels much of that shared scale factor.

Goutham's argument (it's your slide — say it if asked "why keep the weaker measure?"): a **difference score** mathematically cancels the **stable** variance (vascular anatomy, baseline amplitude, trait-level motion) and leaves only what **changed** with the task. So it's **expected** that reconfiguration replicates worse across runs than a raw measure — not a defect that favors activation. Activation replicating better does **not** prove it's a better cognitive measure; it may be loaded with stable (vascular) noise that happens to correlate with performance. With 0-back-alone predicting 0.571, the activation advantage is driven by a **static trait**, not the task.

⚠️ **Traceability gap:** the between-run reliability numbers behind this argument (reconfig ≈ 0.024, activation ≈ 0.169) live in `nb08`, **not** in `pipeline/02`, so they have **no automated check**. **Don't print them on a slide** — keep them verbal, in the speaker note.

Delivery (~1.5–2 min — the longest slide; rehearse against the clock):

> *"Two checks, both of which narrowed our claim. First, direction. At the group level, segregation fell under load — from 0.327 to 0.304, p around 3 times ten-to-the-minus-five, essentially zero by chance. So the direction we predicted is real. But does it predict individuals? The correlation between a person's segregation change and their accuracy is minus 0.105, p = 0.054 — right at the border. So a rock-solid group effect does not give us individual prediction — same trap as blood pressure rising with age in a population but age alone not diagnosing a patient. Second, specificity — naming the axes: horizontal is cross-validated correlation, and each row is a different feature set. 0-back FC, 0.274. Our reconfiguration, 0.366. Both combined, 0.333 — lower, so combining doesn't help. And plain regional activation, 360 features, 0.600 — higher than any connectivity. But this is a specificity check, not a competition: it's post hoc, 360 versus 78 features unmatched, and 0-back activation alone already predicts 0.571 — so it isn't even load-specific. Activation is raw BOLD amplitude, which vascular reactivity contaminates and we can't control here. A difference score like ours mathematically removes that stable variance, so it's expected to look less reliable — that's why we keep it as our pre-specified, amplitude-independent measure."*

Transition: *"So — what does all this add up to?"* (to the conclusion).

Likely questions: - *"So activation is better — should you use it?"* → "It predicts more in this unmatched, post-hoc comparison, but it isn't load-specific and it's confounded by vascular reactivity. We report it as a specificity check, not a replacement." - *"Why does the combined one drop?"* → "Adding 0-back FC to reconfiguration doesn't add independent signal and slightly dilutes it — reconfiguration alone is doing the work." - *"What are the squares?"* → "Cross-run generalization: train on one scan run, test on the other. Both measures drop — reliability across runs is weaker than the main estimate for both." (2 runs per person —

that's where this comes from.) - *"Isn't this dFC?"* → "No — this is static FC aggregated per condition, then subtracted. Not dynamic connectivity."

Avoid: saying a model "wins"; calling reconfiguration "the most robust"; printing the nb08 reliability numbers.

Slide 6 — Conclusions: what we predicted and what the evidence did to it · Kerem · keep projected in Q&A

On screen (verbatim): - Title: *"Predictive signal survives; connectivity-specific mechanism remains unresolved"* - **Survives — the pattern hypothesis:** *"A 78-feature FC difference predicts unseen 2-back accuracy."* · *"The model transfers across identity-disjoint same-HCP cohorts (r = 0.398)."* - **Refined — the directional hypothesis:** *"Segregation fell under load, but larger shifts did not predict better performance."* · *"Reconfiguration showed no clear gain beyond 0-back FC."* · *"FC added no clear gain over activation under the current unmatched comparison."* - **Unresolved:** *"Is the predictive information connectivity-specific, or shared with task activation?"* · *"Vascular reactivity (CVR) is not controlled in the activation benchmark."* - No figure. Andrea approved it with no changes.

What to know: three columns that **close the loop opened on slide 2**. Survives = the Pattern prediction. Refined = the Direction prediction. Unresolved = what we honestly don't know. Naming the first two columns with slide 2's labels is what makes the story **close** rather than sound like a retraction. The 0.398 printed under *Survives* is your closing headline — land it as *"identity-disjoint same-HCP transfer"*, never "independent".

Delivery (~1 min):

> *"To close the loop we opened. Survives — the pattern hypothesis: a 78-feature FC difference predicts unseen 2-back accuracy, and it transfers across identity-disjoint cohorts at r = 0.398. Refined — the directional hypothesis: segregation did fall under load at the group level, but bigger shifts didn't predict better performance; reconfiguration added no clear gain over single-condition FC; and FC added no clear gain over activation in this unmatched comparison. Unresolved: is the predictive information specific to connectivity, or shared with task activation? And vascular reactivity is uncontrolled. So — the predictive signal survives; the connectivity-specific mechanism remains unresolved. Thank you — happy to take questions."*

Likely questions: this slide **is** your Q&A map — keep it projected. Each column points to a backup: Refined → slides 8 and 9, Unresolved → slide 10 (future work).

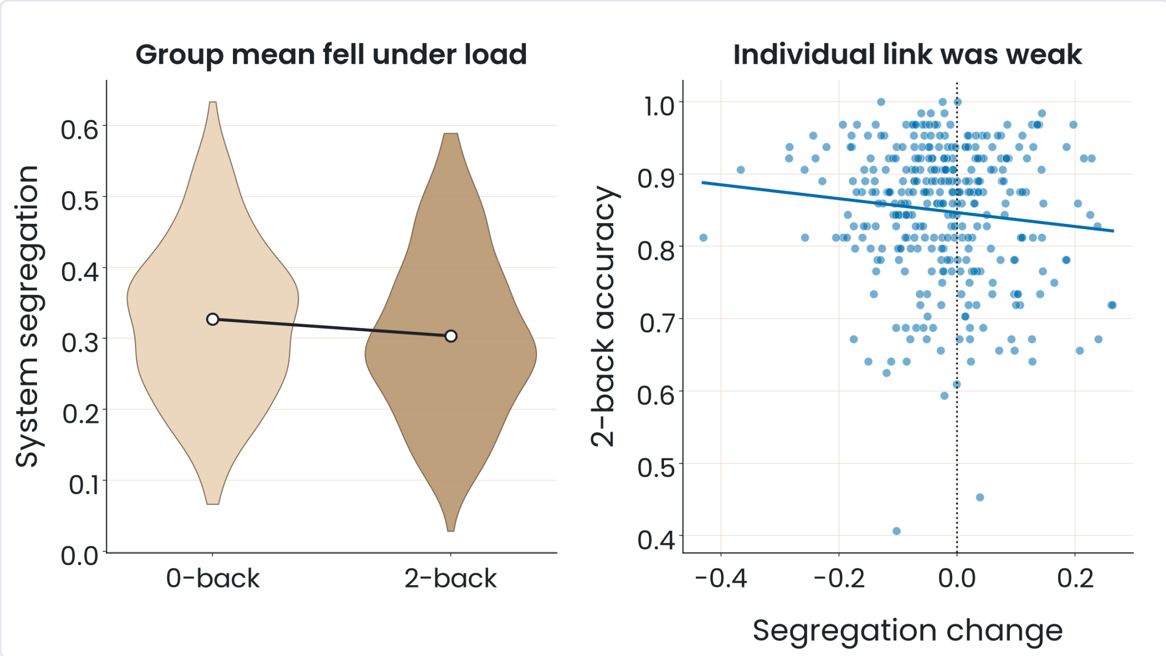
Avoid: saying the conclusion "changed"; that a connectivity mechanism was demonstrated; "independent validation" (it's *identity-disjoint same-HCP*); integration as confirmed at the individual level (group only).

Backups (slides 7–11 · open only if asked)

(Slide 7 = the "Backup · opened only on demand during Q&A" divider. No speaking.)

Slide 8 — Backup: the directional result in full

On screen: Title *"Group direction was real; the individual link was weak"* · *"Group mean: 0-back 0.3271 → 2-back 0.3035"* · *"Paired change: Δ = −0.0236; p = 3.45 × 10^{-5}"* · *"Across participants: r = −0.105; p = .054"* · *"A reliable mean shift does not establish individual predictive relevance."* · Figure `segregation-refinement.png`.



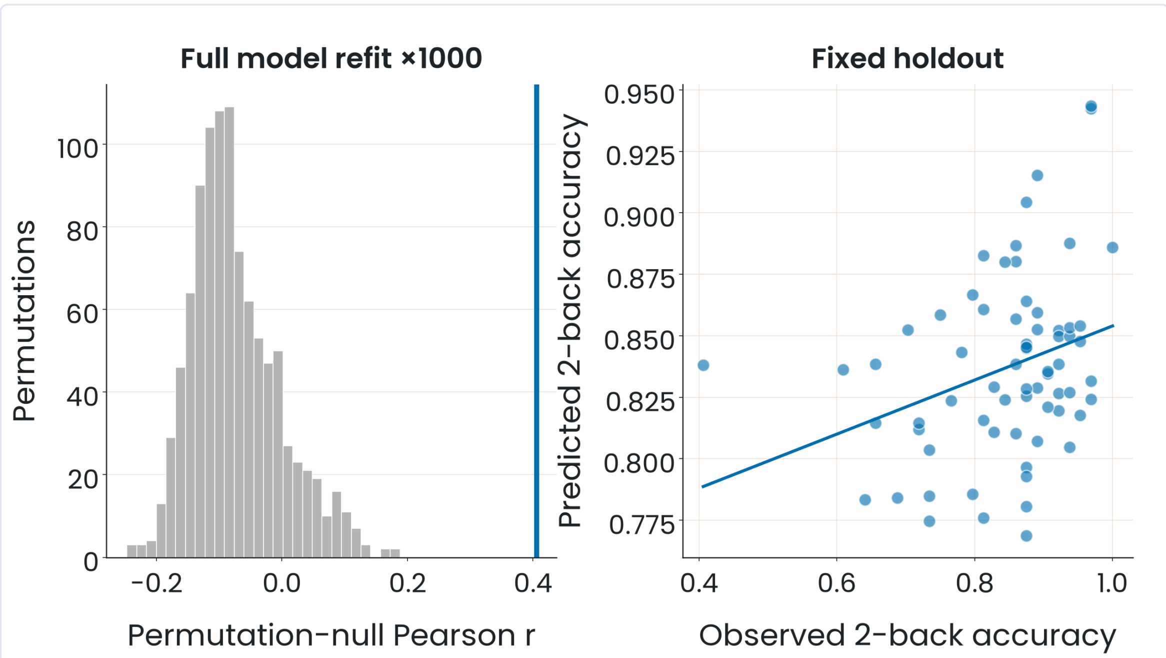
The figure (2 panels): left, two **violins** (0-back / 2-back) = the full segregation distribution of the 336 people, not just the mean; a line with two white dots connects the means and shows the 0.3271 → 0.3035 drop. Right, **scatter**: X = each person's segregation change, Y = their 2-back accuracy; a dotted vertical line at 0 (left = integrated more). The trend is almost flat. **Real group effect, weak individual link** — slide 5's lesson with the full picture.

When to open: "what does 'the directional hypothesis was refined' actually mean?".

Note: don't reuse the **−0.048** magnitude from the submitted abstract — that used a different atlas convention; the canonical one is −0.0236 (same sign, half the magnitude).

Slide 9 — Backup: the checks behind the primary result

On screen: Title *"The checks behind the primary result"* · *"Fixed holdout: r = 0.312 in 67 unseen participants"* · *"Full-refit null (seed 42): r = 0.405; p = 1/1001 ≈ .001"* · *"B → A A-label permutation, fixed B predictions: p = 1/1001 ≈ .001"* · *"Reconfiguration over 0-back FC: ΔR² = +0.0344 ± 0.0225 → no clear gain"* · *"FC over activation: ΔR² = −0.0030 ± 0.0065 → no clear gain"* · *"The permutation p belongs only to seed-42 r = 0.405; the holdout is a separate split."* · Figure `null-and-holdout.png`.



The figure (2 panels): left, **histogram** = shuffle the performance labels 1000 times (refitting the model, fixed partition) → the distribution of "what chance looks like with this pipeline"; a vertical line marks the real r = 0.405, to the right of almost the whole cloud → only 1 of 1001 was that extreme → p ≈ .001. Right, **fixed holdout**: 67 never-touched people, r = 0.312.

⚠️ The one-in-a-thousand p belongs **only** to the seed-42 r = 0.405 — not to the 0.312 holdout nor the 0.366 repeated-CV. Three different procedures, same story, not interchangeable.

The two ΔR² ("no clear gain"): a "2-SD" heuristic (the increment is smaller than twice its own variability). **Not a formal** superiority/equivalence test — say so if asked.

When to open: "how do you know it's not chance / and the holdout?".

Extra ammunition (in `pipeline/02`, not on a slide): we also partial out 0-back accuracy (to rule out "general ability") and repeat with **d′** (signal-detection sensitivity) in cohort B. If asked "did you control for general ability?" → yes.

Slide 10 — Backup: future work

On screen: Title *"Five tests could resolve the remaining question"* · 5 items + decision criterion (*"FC must add reliable held-out value beyond activation and single-condition FC."*).

The 5, what each attacks: (1) match activation/FC dimensionality in nested CV; (2) separate coactivation from coupling with a prespecified estimator; (3) independent-site / repeat-session generalization with kinship modelled; (4) run the activation model on **resting-state** — if rest predicts as well as task, the signal is **task-independent** (runnable now: cohort B ships 4 rest runs); (5) **ST-GNN** (Goutham's proposal, already building it) — model the transition as a dynamic graph and let the model **learn** the reorganization rather than summarize it in one difference ("watch the film, not two snapshots").

When to open: "what next / how would you resolve the activation-vs-FC question?".

Slide 11 — Backup: references and statistical guardrails

On screen: Title *"References and statistical guardrails"* · bibliography (Avery 2020, Chan 2014, Murphy 2020, Masharipov 2024, Hedge 2018, Logothetis 2008) + guardrail definitions (r = 0.366 ± 0.024 = mean ± split SD over 20 partitions; seed-42 r = 0.405, p ≈ .001 = full-refit null; B→A r = 0.398, CI [0.25, 0.53] = identity-disjoint transfer).

How to present it (Andrea): the last presenter shows it for ~5 seconds, says "these are the references we used", and moves to questions. **Don't read it aloud.** Chan backs the segregation formula; Hedge backs Goutham's argument (slide 5); Logothetis frames the CVR limit as a **known** field limitation, not a finding of ours.

Three numbers to memorize (they carry the whole talk): 0.366 ± 0.024 · seed-42 0.405, p ≈ .001 · B→A 0.398, CI [0.25, 0.53].